Conservation Genetics Practice Problems #6

1) Why would an infinite population have no genetic drift?

2) A rare salamander species (diploid, sexually reproducing) loses 1% of its original heterogeneity in a one-generation population crash. Estimate the number of salamanders during this bottleneck.

3) Explain each variable in the formula for the binomial distribution and how it relates to genetic drift over one generation at a biallelic locus.

4) What is the probability an allele at frequency 0.5 will be lost in one generation of random mating in a population of constant size $2N = 4$?

5) Recall the variance of the bionimal distirbution is given by $\sigma = \frac{pq}{2N}$. If genetic drift is maximized when the numerator is greatest, e.g. when $p = q = 0.5$, why are rare alleles more likely to be lost than common alleles?